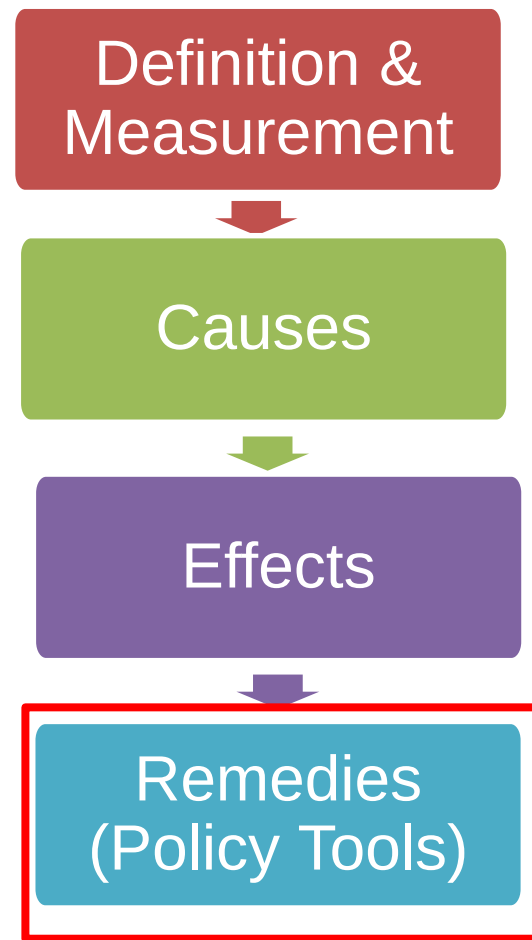
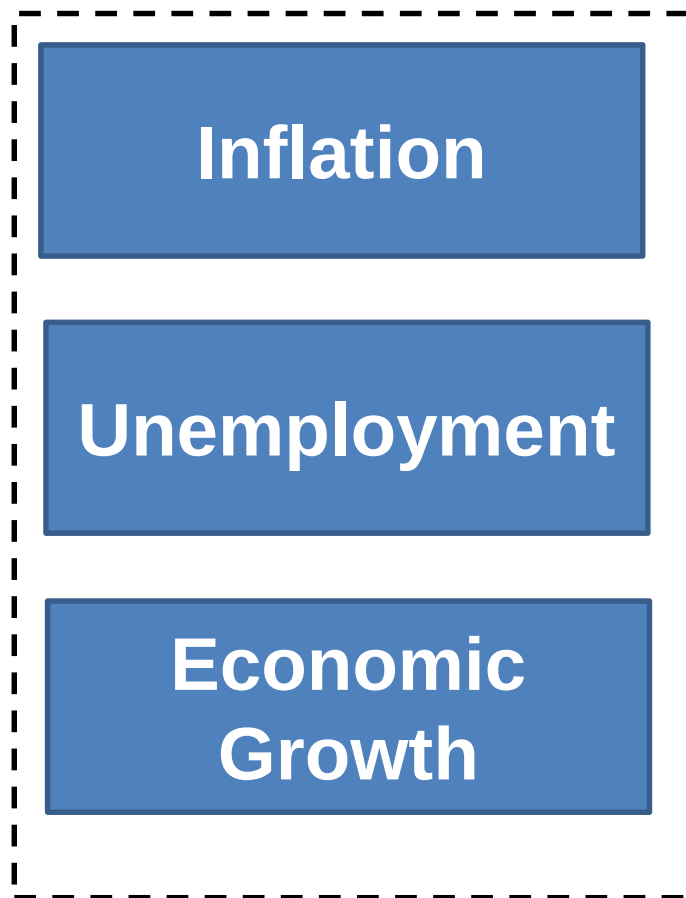




The Big Picture

Macroeconomic Problems



What is inflation?



What is Inflation?

An economic situation when there is a **sustained increase** in the **general price level**.



All prices

vs

Price of one good

Continuous

vs

One Time

What is Inflation?

From textbook (Parry, G. and Kemp, S. 2018)

Inflation is defined as a **persistent** and **appreciable rise** in the **general level of prices** (noticeable price increases that occur over time, and across a range of goods)

What is Inflation?

Households & firms are able to take actions to mitigate the impact of inflation



**Anticipated vs
Unanticipated Inflation**



Results in uncertainty regarding future prices, hence causing undesirable effects on economy



How is inflation measured?

Measuring Inflation

Inflation refers to the rate of change of the general price level (GPL).

The ***inflation rate*** over a period of time is measured by the ***percentage rate of change*** using a given price index.

Rate of inflation =

$$\frac{\text{Price Index in Year 2} - \text{Price Index in Year 1}}{\text{Price Index in Year 1}} \times 100$$

Price Indices

CPI

- Consumer Price Index

RPI

- Retail Price Index

WPI

- Wholesale Price Index

PPI

- Producers Price Index

Price Indices

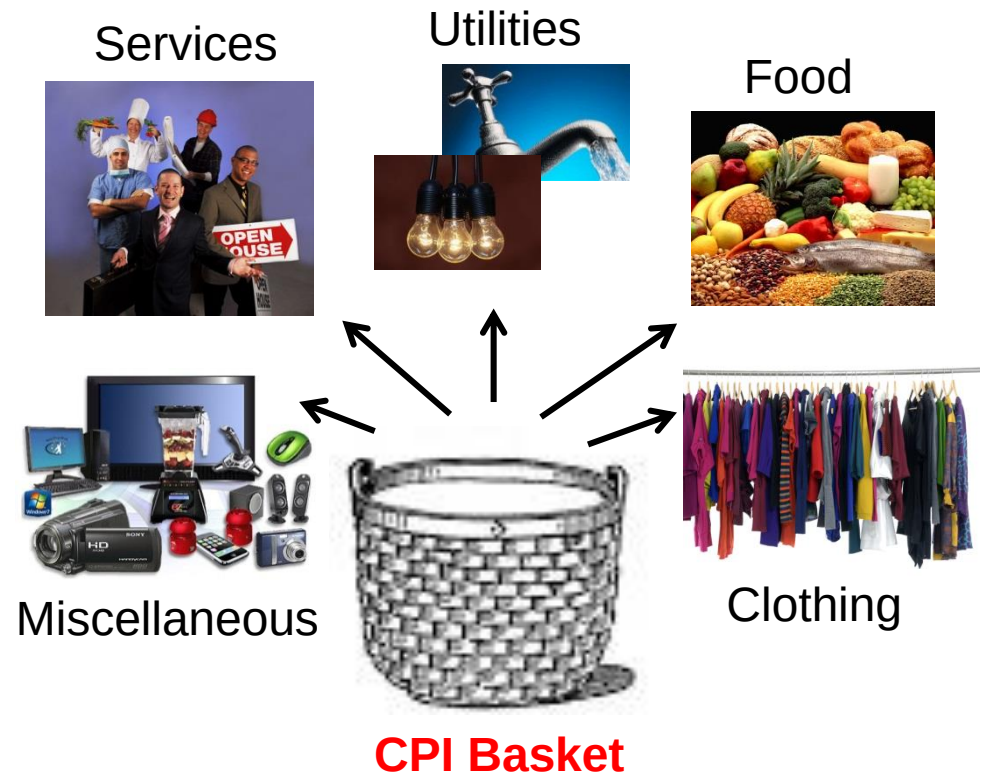
GDP Deflator

- Like the consumer price index (CPI), the GDP deflator is a measure of price inflation/deflation with respect to a specific base year;
- However, unlike the CPI, the GDP deflator is not based on a fixed basket of goods and services; the "basket" for the GDP deflator is allowed to change from year to year with people's consumption and investment patterns.

$$\text{GDP Deflator} = \text{Nominal GDP} / \text{Real GDP} \times 100$$

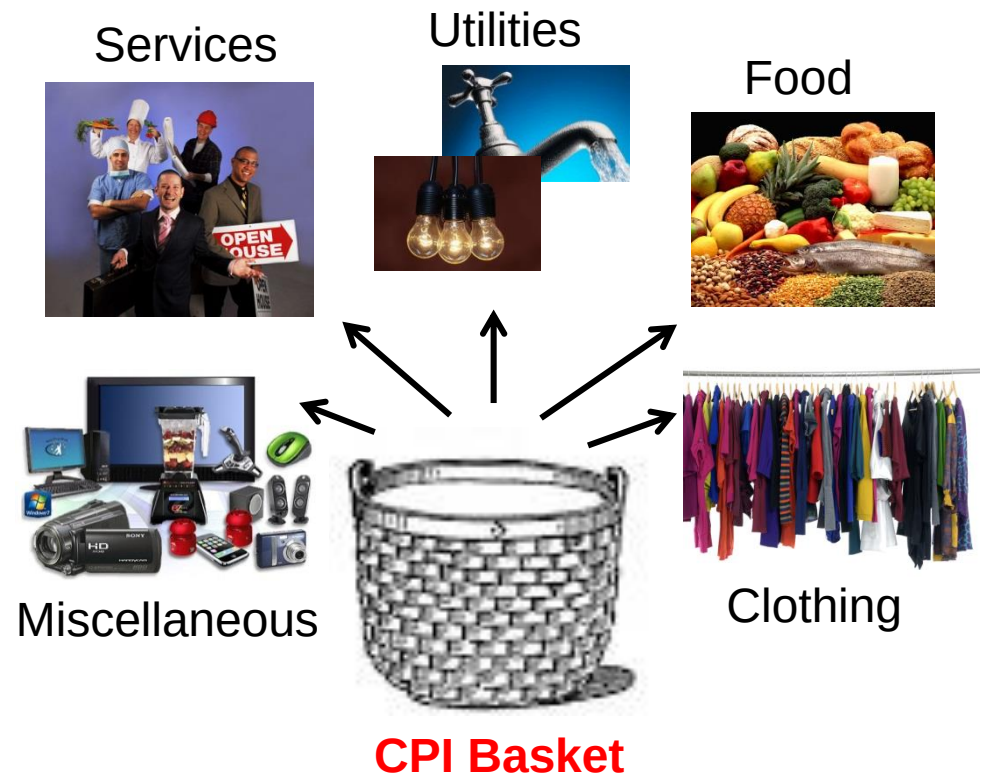
Consumer Price Index (CPI)

- The CPI is a measure that examines the **weighted average of prices** of a **basket of consumer goods and services**.
- It is calculated by taking price changes for each item in the predetermined basket of goods and averaging them.
- The goods are weighted according to their importance.



Consumer Price Index (CPI)

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Consumer Price Index (CPI)

Inflation refers to the rate of change of GPL.

The **inflation rate** between any two periods of time is measured by the **percentage rate of change** in the relevant price index.

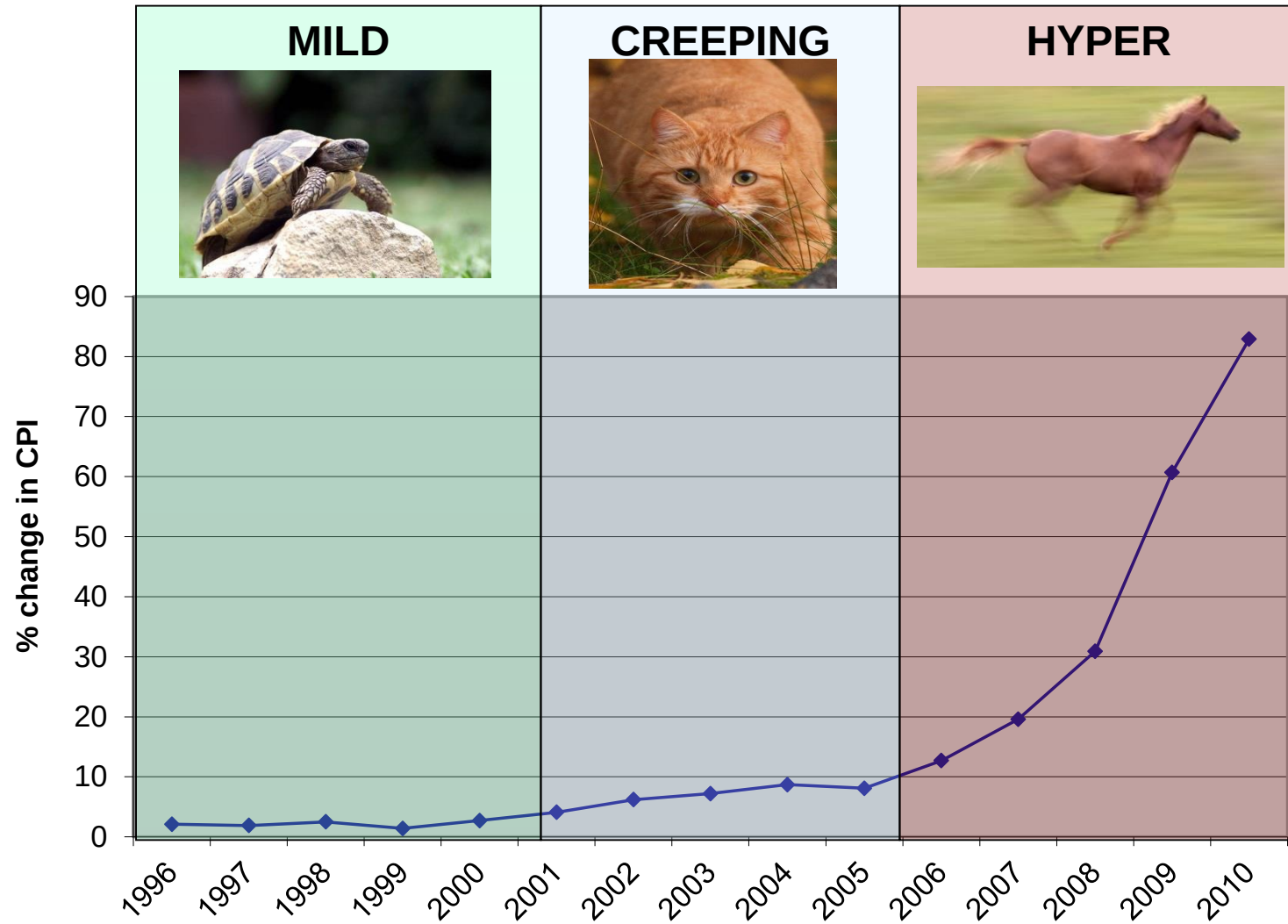
Year-on-Year change in GPL

Year	Price	Index
2009	\$10	100
2010	\$14	140
2011	\$13	130
2012	\$15	150

% change in Price compared to base year	Inflation Rate
40%	40%
30%	-7.1%
50%	15.4%

$$[130 - 140] / 140 \times 100$$

Degrees of Inflation



Example of Hyperinflation: Germany (1921-23)

WHAT

Oct 1923: Monthly inflation rate of 29,500% (Cato Journal)

Dec 1923: 4.2 trillion marks = US\$1.

Hyperinflation had so destroyed the mark that Germans began using it for firewood and coal since the currency was cheaper to burn.

WHY

Germany's decision to fund its war debts by borrowing instead of taxing its citizen's led to a surge in inflation in the Weimar Republic



HOW IT ENDED

The government created an independent central bank and introduced a new currency, the Rentenmark.

Headline and Core Inflation

- **Headline inflation** - is a measure of inflation within an economy, including commodities such as food and energy prices (e.g., oil and gas), which tend to be much more volatile and prone to inflationary spikes.
- On the other hand, "Core inflation" (also non-food-manufacturing or underlying inflation) is calculated from a consumer price index minus the volatile food and energy components.
- Headline inflation may not present an accurate picture of an economy's inflationary trend since sector-specific inflationary spikes are unlikely to persist.

**What are the
causes of inflation?**

Demand-Pull Inflation

Demand-pull inflation occurs
when **aggregate demand (AD)** is
rising at a rate that is not matched by
the **output of goods and services**
(Aggregate Supply)

Demand-Pull Inflation

What causes $\uparrow$ AD leading to rightward shift of AD curve?

$$AD = C + I + G + (X - M)$$

- Rise in consumption expenditure by households
- Rise in firms' investment expenditure
- Rise in demand for country's exports
- Excessive speculative spending
- Rise in government expenditure or decrease tax
- Expansionary fiscal or monetary policy

Cost-Push Inflation

Cost Push Inflation occurs when rising production costs are passed on to consumers who then pay higher prices for final goods and services.

Costs reflect the prices paid for productive inputs.

Periods of cost push inflation can be attributed to increases in prices of significant productive inputs.

Cost-Push Inflation

Cost-push inflation originates from the **supply-side (AS)**

and is caused by a **persistent rise in cost of production (CoP)**

due to **increasing factor prices** (eg. increase in wages, rents, interest etc.)

independent of aggregate demand (AD).

Cost-Push Inflation

Wage Push Inflation

- Caused by persistent $\uparrow$ in wages without corresponding $\uparrow$ in labour productivity.

Import Price-Push Inflation

- Caused by rising import prices resulting in rise in imported consumer goods & raw materials.

Profit Push Inflation

- Caused by firms making use of their market power to make higher profits by raising prices.

Tax Push Inflation

- Caused by increase in taxation.

Exhaustion of Natural Resources

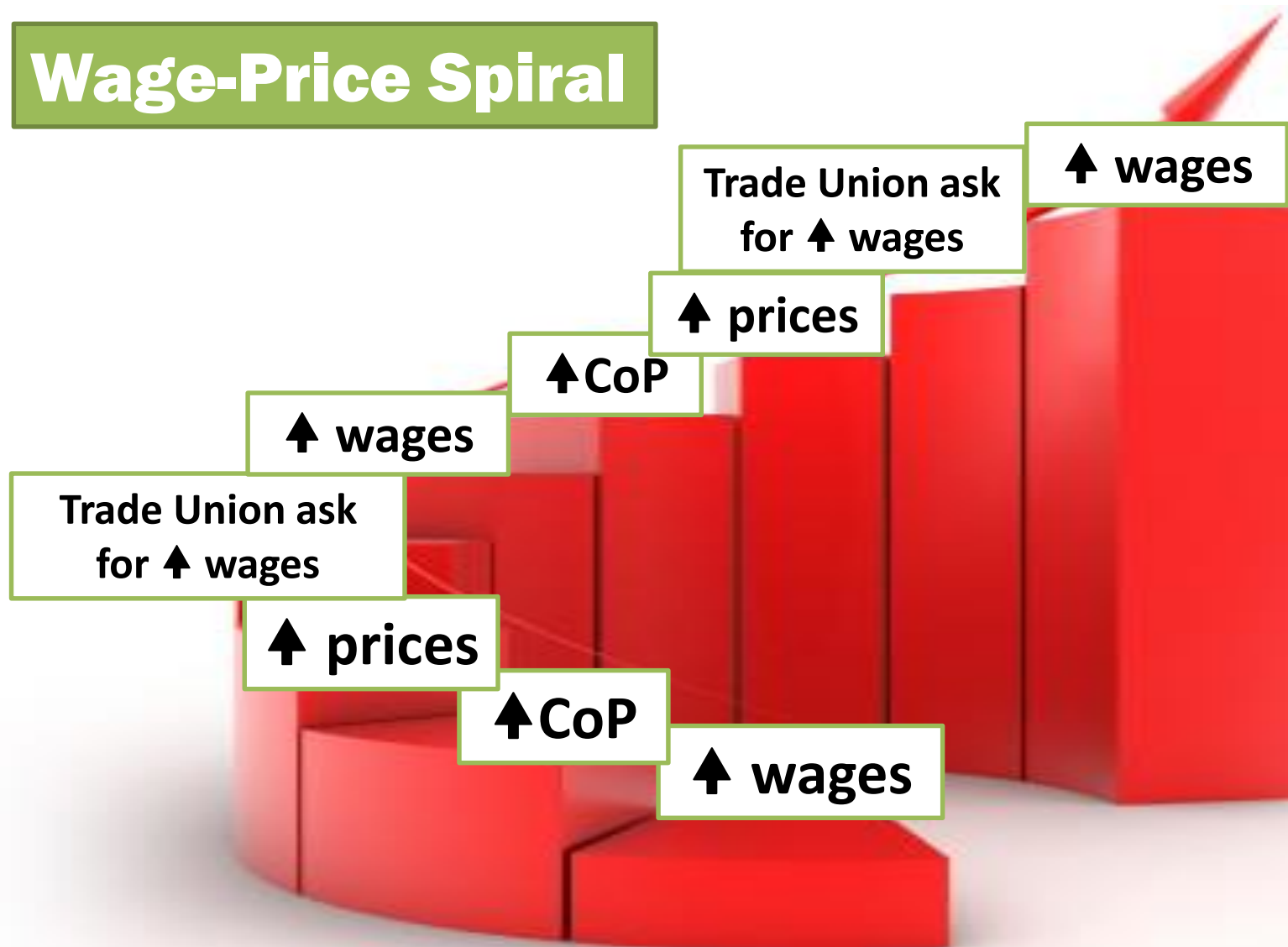
- Caused by depletion of major natural resources for example due to natural disasters.

In Summary

Type of Inflation	Demand-Pull Inflation	Cost-Push Inflation
Causes of Inflation	Caused by persistent increases in AD not matched by output of G&S (AS)	Caused by persistent rises in costs of production, independent of AD
Diagram	Continuous shift of AD curve rightwards, relative to AS	Continuous shift of AS curve upward or leftward, AD curve constant

Interaction of DD-Pull & Cost-Push Inflation

Wage-Price Spiral



What are the effects of inflation?



IMPACTS OF INFLATION

Effects of Inflation

```
graph TD; A[Effects of Inflation] --> B[Internal]; A --> C[External]; B --> D[Loss of confidence in currency]; B --> E[Redistribution of income]; B --> F[Effects on production]; B --> G[Effects on investment]; B --> H[Menu costs]; B --> I[Effects on government finance]; C --> J[Effect on consumers]; C --> K[Effect on producers]; C --> L[Effect on government]; J --> D; J --> E; K --> F; K --> G; L --> I;
```

Internal

→ Loss of confidence in currency

→ Redistribution of income

→ Effects on production

→ Effects on investment

→ Menu costs

→ Effects on government finance

External

Effect on
consumers

Effect on
producers

Effect on
government

Internal Effect:

Loss of Confidence in Currency

- As GPL rises, purchasing power of money falls, *ceteris paribus*.
- Domestic currency will have lower purchasing power $\Rightarrow$ undermine people's confidence in the currency $\Rightarrow$ less willing to hold money.
- Internal value of money falls $\Rightarrow$ people will preserve wealth by buying assets or foreign currencies.
- Domestic currency appreciates (due to lower demand) and assets further increase in price.



Gold Surges as Chinese Inflation Rises

By Dow Jones Business News, July 09, 2013, 05:35:00 AM EDT

Gold prices surged in Europe Tuesday as news that Chinese inflation rose to a four-month high in June helped to boost investors' appetite for the metal, which is seen as a store of value and a hedge against such inflationary pressure.

At 0740 GMT, spot gold was \$1,255.40 a troy ounce, up 1.5% from Monday's price. The metal had earlier traded at its highest price for a week at \$1,260.56 an ounce, while silver rose 1.6% to \$19.360 an ounce.

Internal Effect: Redistribution of Income



Debtors

vs

Creditors



Variable
income earners

vs

Fixed
income earners

Businessmen

vs

Workers

Real savings

vs

Nominal savings

Internal Effect: Redistribution of Income

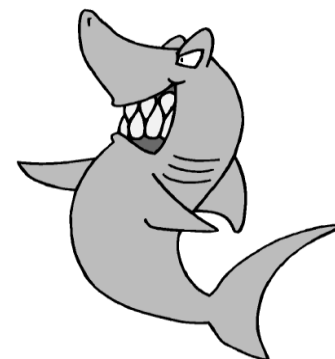


Debtor



- Real value of their debt falls.
- Encourages borrowing.

- Debtor owes creditor S\$100.
- Initially, a S\$100 loan has a purchasing power of S\$100.
- *If GPL rises, what will happen to the purchasing power of S\$100?*
- It will decrease! With a rise in GPL, the S\$100 borrowed will no longer have a purchasing power of S\$100 when the loan is being repaid.



Creditor



- Real value of their loans fall when loans are repaid.
- Discourages lending.

Internal Effect: Redistribution of Income

Variable Income Earner



- Income is linked to GPL.
- As GPL $\uparrow$, nominal income also $\uparrow$. Real value of income remains constant.
- Better off.

Fixed Income Earner



- As GPL $\uparrow$, real value of their incomes $\downarrow$
- Worse off.

Internal Effect: Redistribution of Income



Businessmen



- Since prices will rise faster than costs in the SR, businessmen gain more profits.
- Better off.

Rising prices



Time lag for trade unions to bargain for higher wages and cost of raw materials to rise



In the SR, some costs are fixed (eg. rental)



Workers



- Time lag for bargaining for higher wages.
- Worse off.

Internal Effect: Redistribution of Income

Effects of Inflation on Savings



Internal Effect: Redistribution of Income

Real Savings vs Nominal Savings

- Inflation discourages savings as the real value of savings is eroded
 - But those saving for specific purposes (eg. for retirement or children's education) will have to save more to compensate for the fall in real value of savings.
- Those who save in terms of real assets (eg. jewellery, real estate) gain as the value of these assets tend to rise with rising prices.

**Savers
(in cash) are
worse off**

**Savers
(in assets)
are better off**

Internal Effect:

Effects on Production

Mild Inflation:
Input costs rise slower
than product price

 Stimulates production due to high expected returns
 $\Rightarrow \uparrow I \Rightarrow \uparrow$ employment & productive capacity $\Rightarrow \uparrow Y$

 $\uparrow$ prices $\Rightarrow \uparrow$ profits $\Rightarrow$
 $\uparrow$ complacency &
 $\downarrow$ efficiency $\Rightarrow$ no
incentive to conduct R&D

Creeping / Hyperinflation:
Input costs rise faster
than product price

 Business operate at a
loss $\Rightarrow$ close down $\Rightarrow$
 $\downarrow$ production

 Surviving firms forced
to be more efficient in
order to remain in
business

Internal Effect: Effects on Investment

High inflation rates often associated with uncertainty about the future



Makes planning difficult. Why?



Adverse effect on planned capital investments



Negative effect on LR economic growth and productive capacity
Why?

Internal Effect:

Effects on Government Finance (Bonds**)**

Sells bonds & pays the public a fixed interest rate

**How does the
government
borrow from
the public**



Government



Public

Public buys the bonds & receives interest rate as income

Internal Effect:

Effects on Government Finance

If bonds are sold at
S\$100 each @ 10%
interest rate



Govt pays public
S\$10 per bond as
interest (plus S\$100
when bond matures)

If GPL is rising, the **real cost** of paying the interest **falls** since rate is fixed at time of issue.



Government gains

Internal Effect:

Effects on Government Finance

Direct taxes come from incomes and profits.

- During mild inflation $\Rightarrow \uparrow$ GPL $\Rightarrow \uparrow$ incomes and profits $\Rightarrow \uparrow$ tax receipts $\Rightarrow \uparrow$ government revenue.
- If government revenue rise faster than government expenditure $\Rightarrow$ **budget surplus**
- **In the longer run**, benefits cannot offset the cost of inflation on society and the economy.

Effects

Please Read Textbook Too

Effects of Inflation

```
graph TD; A[Effects of Inflation] --> B[Internal]; A --> C[External]; B --> B1[Loss of confidence in currency]; B --> B2[Redistribution of income]; B --> B3[Effects on production]; B --> B4[Effects on investment]; B --> B5[Effects on government finance]; B --> B6[Menu costs]; C --> C1[Effects on balance of trade]; C --> C2[Effects on capital balance]; C --> C3[Effects on balance of payments & external value of currency]; C --> C4[Costs of foreign debt servicing];
```

Internal

→ Loss of confidence in currency

→ Redistribution of income

→ Effects on production

→ Effects on investment

→ Effects on government finance

→ Menu costs

External

Effects on balance of trade

Effects on capital balance

Effects on balance of payments
& external value of currency

Costs of foreign debt servicing

Food for Thought:
Is Inflation Necessarily Bad?

Why the Fed Worries Inflation Is too Low

By Peter Coy on May 01, 2013

<http://www.businessweek.com/articles/2013-05-01/why-the-fed-worries-inflation-is-too-low>

BloombergBusinessweek
Global Economics

The Federal Reserve's rate-setters announced on Wednesday that they are forging ahead with ultra-easy money, in part because inflation is running "somewhat below" the Fed's target of 2 percent. The Federal Open Market Committee said it will leave the funds rate where it is, just above zero, and keep buying bonds at a pace of \$85 billion a month.

This might strike some readers as odd. Why does the Fed want prices to go up? Isn't inflation bad? And even if you buy the idea that a little inflation is good, who *says* inflation is running under 2 percent? It certainly doesn't feel that way to families that are having trouble making ends meet.

First, the Fed's policymakers want prices to go up because they believe that a little bit of inflation is good for growth. The Fed wants interest rates to be below the rate of inflation to give homeowners and businesses a strong incentive to borrow and spend, generating jobs. The Fed can't do that if there's no inflation because interest rates can't be lower than zero.

Food for Thought: Is Inflation Necessarily Bad?

**The effects of inflation can be good or bad.
It ultimately depends on ...**

- anticipated vs unanticipated inflation
- the rate of inflation: mild, creeping, hyperinflation
- duration of inflation: short or prolonged
- whether productivity or efficiency rises with an increase in price levels.

